

Sem 8

• $x \mid y = \text{NAND} = \overline{xy}$

• $x \downarrow y = \text{NOR} = \overline{x+y}$

1) a) $\overline{(x+y)} \downarrow (xy) = 1$

$$\overline{(x+y)} + (xy) = 1 \Leftrightarrow \overline{(x+y)} + (xy) = 0$$

$\Leftrightarrow \overline{(x+y)} = 0 \quad xy = 0$

$x+y = 1$

~~(1;1)~~; (0;1); (1;0)

(0;1); (1;0)

$$(x; y) = (0; 1), (1; 0)$$

b) $(x \mid y) + (\bar{x}y) = 0$

$$\overline{(xy)} + (\bar{x}y) = 0$$

$$\overline{(xy)} = 0$$

$$xy = 1 \Leftrightarrow (1; 1)$$

$$(\bar{x}y) = 0$$

~~(0;0)~~; ~~(1;0)~~; (1;1)

$$(x, y) = (1, 1)$$

2) a) $F(x, y, z) = (x+y)\bar{z} + (x|y)$

X	y	Z	$x+y$	$\bar{z}$	$\overline{xy}$	$(x+y)\bar{z}$	$(x+y)\bar{z} + (x y)$
1	1	1	1	0	0	0	0
1	1	0	1	1	0	1	1
1	0	1	1	0	1	0	1
1	0	0	1	1	1	1	1
0	1	1	1	0	1	0	1
0	1	0	1	1	1	1	1
0	0	1	0	0	1	0	1
0	0	0	0	1	1	0	1

b) $F(x, y, z) = x\bar{y} + z + (x \downarrow y)$

X	y	Z	$\bar{y}$	$x\bar{y}$	$\overline{(x+y)}$	$x\bar{y} + z$	$x\bar{y} + z + (x \downarrow y)$
1	1	1	0	0	0	1	1
1	1	0	0	0	0	0	0
1	0	1	1	1	0	1	1
1	0	0	1	1	0	1	1
0	1	1	0	0	0	1	1
0	1	0	0	0	0	0	0
0	0	1	1	0	1	1	1
0	0	0	1	0	1	0	1

Nº 3

Find the sum-of-products expansions and product-of-sums expansions of these Boolean functions.

a) $F(x, y) = x + \bar{y} + (x \downarrow y)$

x	y	$\bar{y}$	$\overline{x+y}$	$x + \bar{y} + (x \downarrow y)$
1	1	0	0	① ← DNF
1	0	1	0	①
0	1	0	0	0 ← CNF
0	0	1	1	①

Sum-of-products (DNF)
 $xy + x\bar{y} + \bar{x}\bar{y}$

$$(x \downarrow y) = \overline{(x+y)} = \bar{x}\bar{y}$$

De Morgan's law
 $\overline{(x+y)} = \bar{x}\bar{y}$
 $\overline{(xy)} = \bar{x} + \bar{y}$

Product-of-sums (CNF)
 $(x + \bar{y})$

b) $F(x, y, z) = xy + \bar{x}yz + (x \downarrow y)$

Sum-of-products (DNF)

~~$x\bar{y}z + \bar{x}yz + x\bar{y}z$~~ $xyz + xy\bar{z} + xz\bar{y} + yz\bar{x} + z\bar{x}\bar{y}$

Product-of-sums (CNF)

~~$\bar{x}\bar{y}z + z\bar{x}\bar{y} + yz\bar{x} + xz\bar{y} + x\bar{y}z$~~ ~~$y\bar{z}\bar{x} + xz\bar{y} + x\bar{y}z$~~
 $(x+y+z)(x+z+\bar{y})(y+z+\bar{x})$

	x	y	z	xy	$\overline{xy}$	$\overline{xy}z$	$\overline{xy}z (x y)$	$xy + \overline{xy}z (x y)$
1	1	1	1	1	0	0	0	1
1	1	1	0	1	0	0	0	1
1	1	0	1	0	1	1	1	1
1	1	0	0	0	1	0	0	0
1	0	1	1	0	1	1	1	1
1	0	1	0	0	1	0	0	0
1	0	0	1	0	1	1	1	1
1	0	0	0	0	1	0	0	0

№4

Find DNF and CNF of the Boolean function $F(x, y, z)$ that equals 1 if and only if $xy + \bar{z} = 0$.

x	y	z	$\bar{z}$	xy	$xy + \bar{z}$	$f(x, y, z)$
1	1	1	0	1	1	0
1	1	0	1	1	1	0
1	0	1	0	0	0	1
1	0	0	1	0	1	0
0	1	1	0	0	0	1
0	1	0	1	0	1	0
0	0	1	0	0	0	1
0	0	0	1	0	1	0

DNF: $xz\bar{y} + yz\bar{x} + z\bar{x}\bar{y}$

CNF: $(x+y+z)(x+z+\bar{y})(y+z+\bar{x})(z+\bar{x}+\bar{y})(\bar{x}+\bar{y}+\bar{z})$

N₅

$$a) (x|y) + xy \Leftrightarrow \overline{xy} + xy = 1 \Leftrightarrow \overline{\overline{xy} + xy}$$

$$(x|y) + xy = \overline{\overline{xy}}$$

$$\overline{\overline{xx}} = 1$$

$$\overline{\overline{xx}} = 0$$

$$b) xy + \overline{(y+z)} \Leftrightarrow xy + \overline{y} \overline{z} \Leftrightarrow \overline{\overline{xy} \cdot \overline{y} \overline{z}}$$

$$A+B = \overline{\overline{A} \cdot \overline{B}}$$

N₆

$$x \downarrow y = \overline{x+y} \Leftrightarrow \overline{x \downarrow y} = (x \downarrow y) \downarrow (x \downarrow y) = x+y$$

$$x \downarrow x = \overline{x+x} = \overline{x} \Leftrightarrow \overline{x} = x \downarrow x$$

$$x+y = (x \downarrow y) \downarrow (x \downarrow y)$$

$$xy = \overline{\overline{x} + \overline{y}}$$

$$xy = (x \downarrow x) \downarrow (y \downarrow y)$$

N₇

All functions obtained from + and . are monotone, because both OR and AND are monotone operations. But the Boolean function $\overline{x}$ is not monotone, since $0 < 1$ but $\overline{0} = 1$ and $\overline{1} = 0$.

Hence $\overline{x}$ cannot be expressed using only + and . . Therefore $\{+, \cdot\}$ is not functionally complete.